

Nathan Liu

Danville, CA | nathan.dtliau@gmail.com | [linkedin.com/in/n8liu](https://www.linkedin.com/in/n8liu) | nliu.org | (925) 786-3685

EDUCATION

University of California, Berkeley

Bachelors of Arts in Data Science & Computer Science

Berkeley, CA

August 2026

TECHNICAL SKILLS

Languages: Python, Java, SQL (PostgreSQL), C/C++, TypeScript

Developer Tools: Git, Docker, Kubernetes, AWS (DynamoDB, CloudWatch, S3, RDS, EC2)

Frameworks: Django, FastAPI, React, Node.js, PySpark

Libraries: pandas, NumPy, Matplotlib, Pytorch

PROFESSIONAL EXPERIENCE

Data Engineer Intern

Carbon Sustain

May 2025 – August 2025

Berkeley, CA

- Engineered a real-time event-processing pipeline handling 100K+ events/day, adding schema validation and S3 execution traces to support deterministic replay and faster incident debugging.
- Enhanced system reliability by implementing DynamoDB idempotency keys to handle retry spikes and configuring CloudWatch alarms, cutting real-time execution failures by 62%.
- Analyzed catering and digital-media footprint data to identify decarbonization opportunities representing up to 76% of modeled event emissions.

Software Engineer

UnifIBD

January 2025 – May 2025

Remote (San Jose, CA)

- Reduced time-to-production for data features by half by standardizing analytics-to-engineering integration workflows and implementation handoffs.
- Removed manual mobile releases to TestFlight and Google Play with Fastlane and GitHub Actions, reducing deployment time from hours to minutes.
- Led a three-engineer migration of a Swift iOS application to Flutter, defining the architecture and technical milestones while consolidating platform-specific code into a shared codebase.

Data Science Intern

L.A. Lucky Import & Export Inc.

June 2024 – August 2024

Los Angeles, CA

- Developed statistical forecasting pipelines and inventory anomaly detection across 4,000+ SKUs, identifying seasonal surges to improve supply chain and reduce inventory shrinkage by 60%.
- Reduced API response latency by 78% by minimizing response payloads and introducing server-side caching for frequently accessed inventory data.
- Reconstructed a real-time SAP inventory sync using incremental pulls and upserts into PostgreSQL which reduced manual processing substantially and enabled automated invoicing for all customers.

PROJECTS

CardboardDex | TypeScript, Python, PostgreSQL

2026

- Built a production TCG price tracker full pipeline from eBay listings through a custom NLP title matching, served via FastAPI + Next.js on a hybrid Cloudflare + AWS stack with automated CI/CD.
- Engineered high-performance search over 66,000+ cards using multi-tier caching achieving sub-5ms queries against a live AWS RDS instance tracking 110,000+ price observations.

FleetManager | Python, Rust, PostgreSQL, AWS

2025

- Developed predictive risk-scoring models using driver behavior and vehicle-utilization signals to prioritize preventive-maintenance alerts.
- Architected and debugged production REST APIs in Django and PostgreSQL, supporting bulk telemetry ingestion to AWS S3, utility boundary resolution, and predictive vehicle trip modeling.